Day 3 – DAC Characterization

Session Speaker:

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DAC CHARATERIZATION

- For a **4-bit voltage-output R–2R DAC modelled in Cadence Virtuoso**, you can characterize it in two groups:
  - **Parametric Transient Analysis for Static Characterization**.
  - **Coherent Transient Analysis followed by an FFT (or Spectrum Analysis)**.

- For a 4-bit DAC, there are only:

$$2^4 = 16$$

- input codes:

$$0000 \rightarrow 1111$$

PARAMETERS

No.	Parameter	What it tells you	How to calculate
1	Resolution	Number of bits	$N = 4$ bits
2	LSB	Ideal voltage step	$V_{REF}/16$
3	Offset error	Error near zero code	Actual output at code 0 – ideal output
4	Gain error	Error in overall slope	Compare actual and ideal full-scale/slope
5	DNL	Error in each code step	$\frac{V_{k+1} - V_k}{LSB} - 1$
6	INL	Deviation from ideal transfer curve	$\frac{V_k - V_{ideal,k}}{LSB}$
7	Monotonicity	Whether output always increases	$V_{k+1} \geq V_k$
8	Full-scale error	Error at maximum code	Actual FS – ideal FS
9	Settling time	How quickly output reaches final value	Time to remain within ±0.5 LSB
10	Rise time	Output transition speed	$t_{90} - t_{10}$
11	Fall time	Falling transition speed	$t_{10} - t_{90}$
12	SNDR	Signal quality	FFT
13	SNR	Signal/noise ratio	FFT
14	THD	Harmonic distortion	FFT
15	SFDR	Largest spur relative to fundamental	FFT
16	ENOB	Effective resolution	$(SNDR - 1.76)/6.02$

Ideal LSB

For a 4-bit DAC:

$$2^4 = 16$$

If your reference voltage is, for example,

$$V_{REF} = 1.8V$$

then:

$$LSB = \frac{V_{REF}}{2^4}$$

$$LSB = \frac{1.8}{16}$$

$$\boxed{LSB = 112.5mV}$$

Your ideal output is:

- $V_{OUT,ideal}$ = (V_{REF} * Code)/16

IDEAL OUTPUT TABLE

Decimal	D3 D2 D1 D0	Ideal V_{OUT}@ 1.8 V
0	0000	0 V/0.29
1	0001	0.1125 V
2	0010	0.2250 V
3	0011	0.3375 V
4	0100	0.4500 V
5	0101	0.5625 V
6	0110	0.6750 V
7	0111	0.7875 V
8	1000	0.9000 V
9	1001	1.0125 V
10	1010	1.1250 V
11	1011	1.2375 V
12	1100	1.3500 V
13	1101	1.4625 V
14	1110	1.5750 V
15	1111	1.6875 V

The maximum code is 15, not 16.
Therefore:

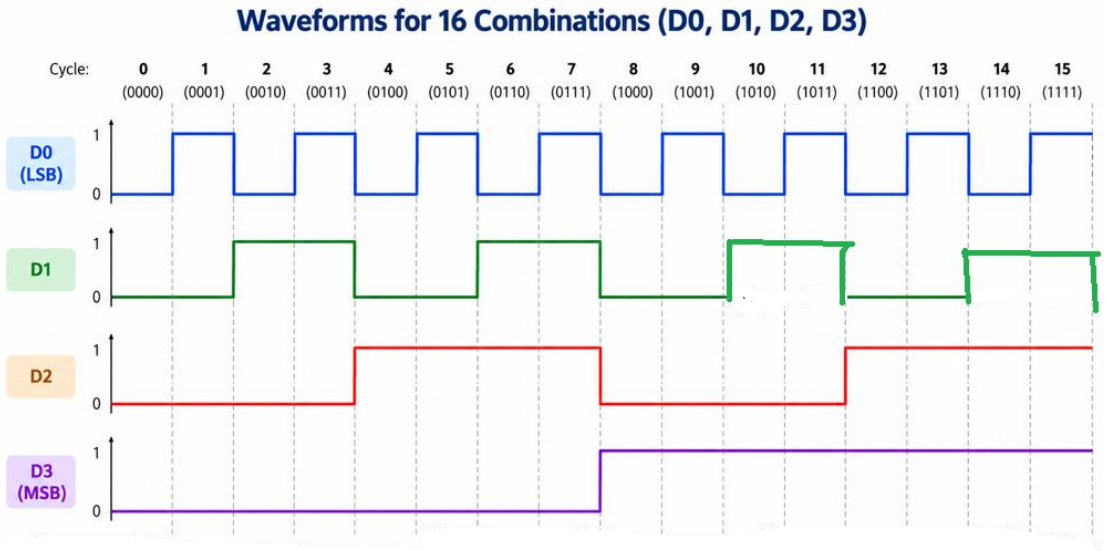
$$V_{FS,ideal} = V_{REF}\frac{15}{16}$$

$V_{FS,ideal}$=1.6875V

DC TRANSFER CHARACTERISTIC

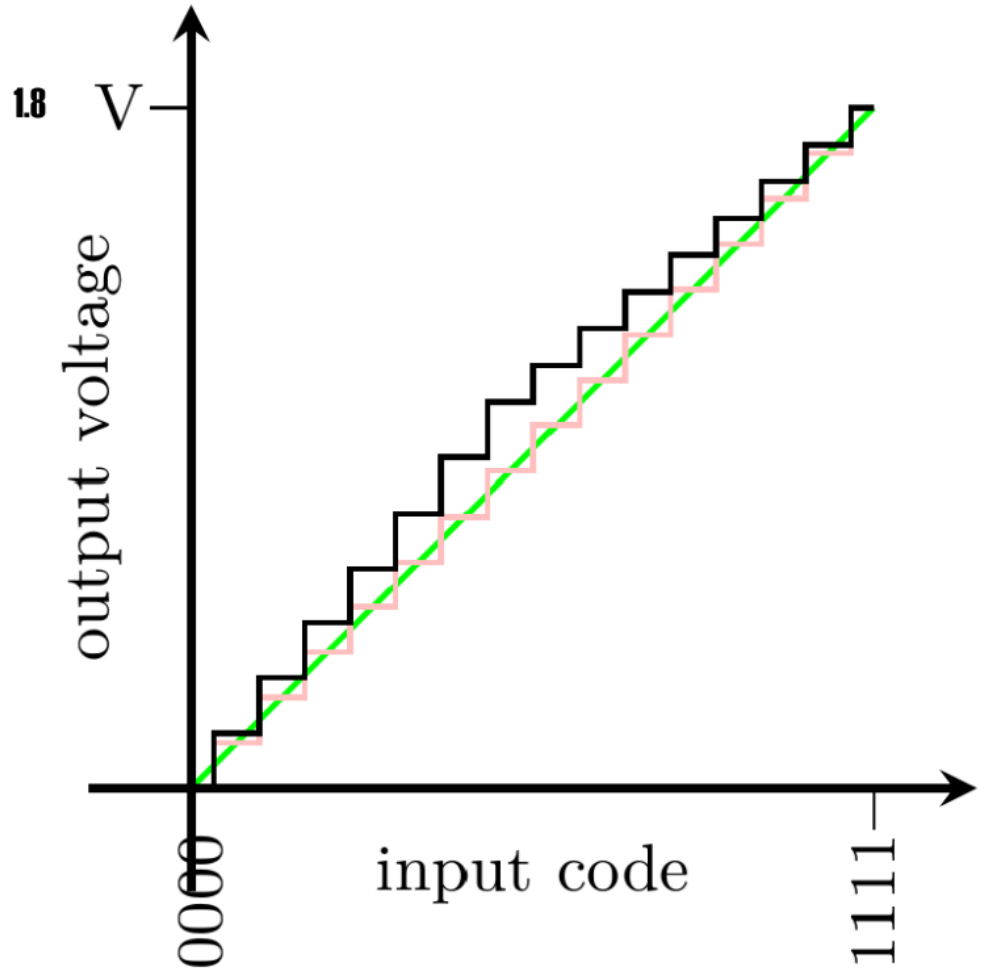
- Measure V_{OUT} for every code and plot: V_{OUT} vs Digital Code
- This single simulation gives you the information needed for **DNL, INL, monotonicity, gain error and offset error**.
- DNL/INL Glitches:
  - At major transitions (especially from 0111 to 1000), you might see sharp voltage spikes or glitches. This is normal in practical R-2R or binary-weighted DAC designs due to minor timing mismatches between bits turning on and off.
  - Settling Time: Check if the output completely flattens out before the next input transition occurs. If it doesn't settle, increase your pulse widths (e.g., scale everything up to milliseconds instead of microseconds).

TEST-BENCH SETUP



Parameter	D_0 (LSB)	D_1	D_2	D_3 (MSB)
Voltage 1	0	0	0	0
Voltage 2	VDD (e.g., 1.2)	VDD	VDD	VDD
Delay time	0	0	0	0
Rise time	1n	1n	1n	1n
Fall time	1n	1n	1n	1n
Pulse width	1u	2u	4u	8u
Period	2u	4u	8u	16u

DC TRANSFER CHARACTERISTIC



Measure V_{OUT} for every code and plot: V_{OUT} vs Digital Code

DNL — DIFFERENTIAL NON-LINEARITY

DNL tells you whether each actual DAC step is equal to one ideal LSB.

For code k:

$$\boxed{DNL(k) = \frac{V_{OUT}(k+1) - V_{OUT}(k)}{LSB_{ideal}} - 1}$$

Suppose:

$$V_{OUT}(5) = 0.558V$$

and

$$V_{OUT}(6) = 0.670V$$

Then:

$$Actual_Step = 0.670 - 0.558$$
$$= 0.112V$$

Ideal:

$$LSB = 0.1125V$$

Therefore:

$$DNL(5) = \frac{0.112}{0.1125} - 1$$

$$\boxed{DNL(5) \approx -0.0044\,LSB}$$

For a 4-bit DAC you will have **15 DNL values**, because there are 15 transitions between 16 codes.

DNL CALCULATION

Decimal	Binary	Ideal (V)	Practical V_{OUT}(V)
0	0000	0.0000	0.003
1	0001	0.1125	0.114
2	0010	0.2250	0.226
3	0011	0.3375	0.340
4	0100	0.4500	0.448
5	0101	0.5625	0.565
6	0110	0.6750	0.673
7	0111	0.7875	0.790
8	1000	0.9000	0.896
9	1001	1.0125	1.015
10	1010	1.1250	1.124
11	1011	1.2375	1.241
12	1100	1.3500	1.347
13	1101	1.4625	1.465
14	1110	1.5750	1.571
15	1111	1.6875	1.691

Transition	V_k	V_{k+1}	Actual step	DNL
0000→0001	0.003	0.114	0.111	−0.0133
0001→0010	0.114	0.226	0.112	−0.0044
0010→0011	0.226	0.340	0.114	+0.0133
0011→0100	0.340	0.448	0.108	−0.0400
0100→0101	0.448	0.565	0.117	+0.0400
0101→0110	0.565	0.673	0.108	−0.0400
0110→0111	0.673	0.790	0.117	+0.0400
0111→1000	0.790	0.896	0.106	−0.0578
1000→1001	0.896	1.015	0.119	+0.0578
1001→1010	1.015	1.124	0.109	−0.0311
1010→1011	1.124	1.241	0.117	+0.0400
1011→1100	1.241	1.347	0.106	−0.0578
1100→1101	1.347	1.465	0.118	+0.0489
1101→1110	1.465	1.571	0.106	−0.0578
1110→1111	1.571	1.691	0.120	+0.0667

DNL_{max}=+0.0667LSB

$$\boxed{DNL_{min} = -0.0578LSB}$$

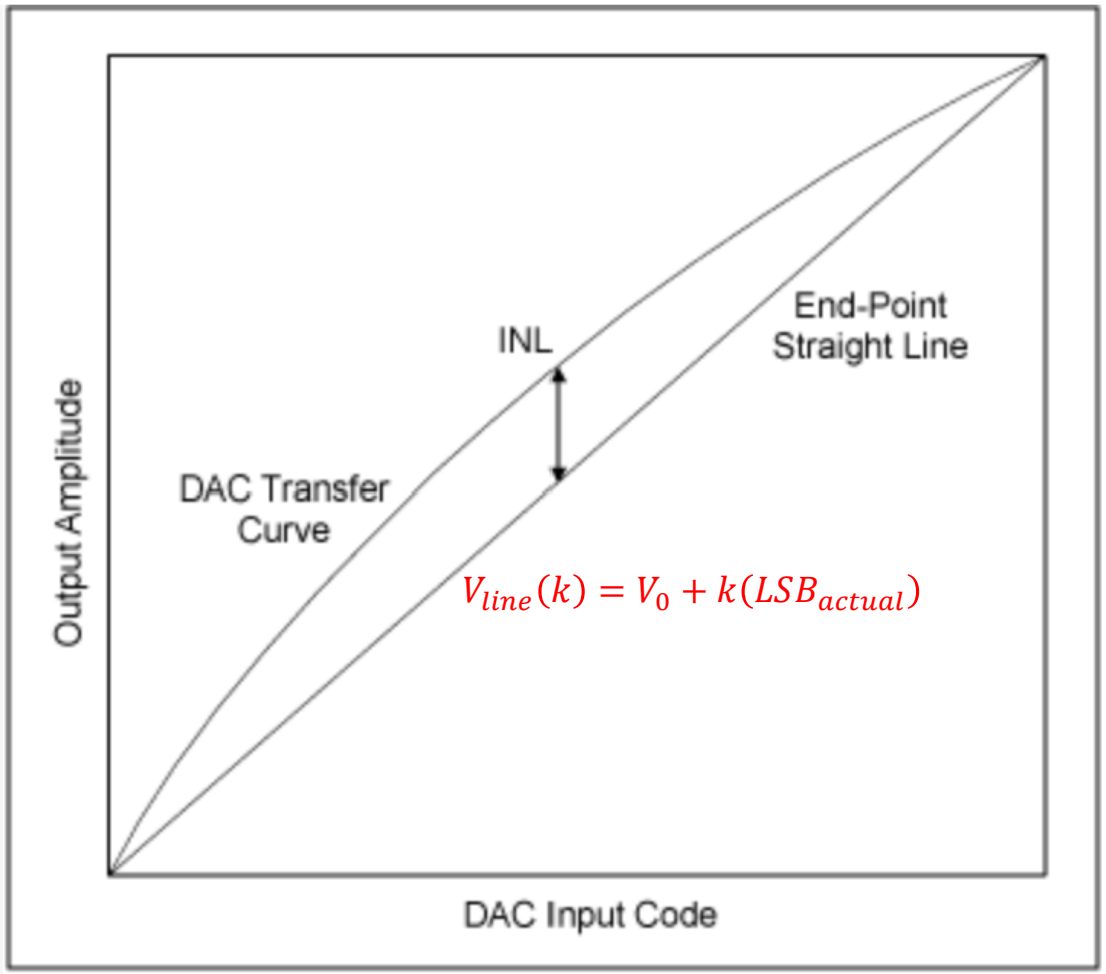
Worst-case DNL:

$|DNL|_{max}$ =0.0667LSB

DNL should be between -1 and +1 LSB (|DNL| ≤ 1 LSB)

INL-INTEGRAL NON-LINEARITY

- Represents the **straight reference line** against which you compare the actual DAC output.



Endpoint INL:

$V_{line}(k) = V_0 + k(LSB_{actual})$ then $INL_k = \frac{V_k - V_{line}(k)}{LSB_{actual}}$

1. First calculate the actual slope using codes 0 and 15.

$$V_0 = 0.003V$$

$$V_{15} = 1.691V$$

Therefore:

$$LSB_{actual} = \frac{V_{15} - V_0}{15}$$

$$= \frac{1.691 - 0.003}{15}$$

$$= \frac{1.688}{15}$$

LSB_{actual} =0.112533V

The endpoint reference line is:

$$V_{line}(k) = V_0 + k(LSB_{actual})$$

Then:

$$INL_k = \frac{V_k - V_{line}(k)}{LSB_{actual}}$$

Actual:

$$V_7 = 0.790V$$

Endpoint line:

$$V_{line,7} = 0.003 + 7(0.112533)$$

$$= 0.79073V$$

Therefore:

$$INL_7 = \frac{0.790 - 0.79073}{0.112533}$$

$$INL_7 = -0.0065\text{LSB}$$

INL should ideally be within ± 0.5 to ± 1 LSB
(|INL| ≤ 1 LSB)

Decimal	Binary	Actual V_{OUT}	Endpoint line (V)	INL (LSB)
0	0000	0.003	0.0030	0.0000
1	0001	0.114	0.1155	−0.0133
2	0010	0.226	0.2281	−0.0187
3	0011	0.340	0.3406	−0.0053
4	0100	0.448	0.4531	−0.0453
5	0101	0.565	0.5657	−0.0062
6	0110	0.673	0.6782	−0.0462
7	0111	0.790	0.7907	−0.0065
8	1000	0.896	0.9033	−0.0649
9	1001	1.015	1.0158	−0.0071
10	1010	1.124	1.1283	−0.0382
11	1011	1.241	1.2408	+0.0018
12	1100	1.347	1.3534	−0.0568
13	1101	1.465	1.4659	−0.0080
14	1110	1.571	1.5781	−0.0630
15	1111	1.691	1.6910	0.0000

INL_{max} ≈+0.0018LSB

$INL_{min} \approx -0.0649LSB$

Worst-case:

$|INL|_{max} \approx -0.0649LSB$

OFFSET ERROR

- Offset error is the deviation of the actual zero-code output from the ideal zero output.

Ideal:

$$V_{ideal}(0000) = 0$$

Actual:

$$V_{actual}(0000) = 0.003V$$

Therefore:

$$\boxed{V_{offset} = 3mV}$$

In LSB:

$$Offset_{LSB} = \frac{0.003}{0.1125}$$

Offset= +0.0267LSB

FULL-SCALE ERROR

At code 1111:

Ideal:

$$V_{ideal,FS} = 1.6875V$$

Actual:

$$V_{actual,FS} = 1.691V$$

Therefore:

$$E_{FS} = 1.691 - 1.6875$$

$$\boxed{E_{FS} = +3.5mV}$$

In LSB:

$$E_{FS,LSB} = \frac{0.0035}{0.1125}$$

EFS=+0.0311LSB

MONOTONICITY

A DAC is monotonic if:

$$V_{k+1} \geq V_k$$

for every code transition.

Decimal	Binary	Actual V_{OUT}
0	0000	0.003
1	0001	0.114
2	0010	0.226
3	0011	0.340
4	0100	0.448
5	0101	0.565
6	0110	0.673
7	0111	0.790
8	1000	0.896
9	1001	1.015
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14	1110	1.571
15	1111	1.691

RISE AND FALL TIME

Apply digital transitions.

For example:

$$0000 \rightarrow 1111$$

Observe:

$$V_{OUT}(t)$$

You can characterize:

Rise time

Measure the time between 10% and 90% of the final output:

$$\boxed{t_r = t_{90} - t_{10}}$$

Fall time

$$\boxed{t_f = t_{10} - t_{90}}$$

for a falling transition.

SETTLING TIME

- The **settling time** is the time from the code transition until the output enters this region and **stays there**.

$$t_{settling} = \text{time to remain within specified error band}$$

DYNAMIC PERFORMANCE USING FFT

- Apply a changing digital code representing a sinusoidal signal and output is analysed in frequency domain.

SNR

$$SNR = 10\log_{10}\left(\frac{P_{signal}}{P_{noise}}\right)$$

SNDR

$$SNDR = 10\log_{10}\left(\frac{P_{signal}}{P_{noise} + P_{distortion}}\right)$$

THD

$$THD = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \cdots}}{V_1}$$

where V_1 is the fundamental and V_2, V_3, …are harmonics.

SFDR

$$SFDR = 20\log_{10}\left(\frac{V_{fundamental}}{V_{largest\ spur}}\right)$$

EFFECTIVE NUMBER OF BITS

- **ENOB** tells you how many bits of resolution your real ADC effectively provides.

$$ENOB = \frac{SNDR - 1.76}{6.02}$$

Because of following parameters:

- Quantization noise
- Thermal noise
- Circuit noise
- Nonlinearity
- Harmonic distortion
- Other imperfections

A nominal **14-bit ADC** might have an ENOB of only **11.8 bits**.

SPURIOUS-FREE DYNAMIC RANGE

- SFDR tells you how small a signal you can detect in the presence of a much larger dominant signal before that small signal gets buried

$$SFDR = 20\log_{10}\left(\frac{V_{fundamental}}{V_{largest\ spur}}\right)$$

COHERENT TRANSIENT ANALYSIS FOLLOWED BY AN FFT

- Apply a changing digital code representing a sinusoidal signal and output is analysed in frequency domain.
- Apply a **continuously changing digital code sequence** so that the DAC output approximates a periodic waveform, commonly a sine wave.

Then:

$$\boxed{\text{Digital code sequence} \rightarrow DAC \rightarrow V_{OUT}(t) \rightarrow FFT}$$

The FFT tells you how much energy is present in:

- Fundamental signal
- Harmonics
- Noise
- Spurious tones

Sinusoidal digital code sequence.

For a DAC:

$$D[n] = round\left[\frac{2^N - 1}{2}\left(1 + A\sin(2\pi f_{in}\, n\, /F_s)\right)\right]$$

$$= round\left[\frac{2^N - 1}{2}\left(1 + \sin\left(2\pi \cdot \frac{M}{N_{samples}} \cdot n\right)\right)\right]$$

For your 4-bit DAC:

$$D[n] = round\left[7.5\left(1 + A\sin(2\pi f_{in}\, n\, /F_s)\right)\right]$$

This generates codes between 0 and 15 following a sine-wave trajectory.

- A: The amplitude scaling factor. If $A = 1$, the sine wave swings fully from 0 to 15.
  - If $A = 0.5$, the wave will only swing half as high.
- f_{in}: The frequency of the desired analog sine wave.
- F_s: The sampling frequency (your digital clock speed).
- n: The current sample number (0,1,2,3, …).
- $f_{in} = \frac{M}{N_{samples}} \times F_s$



- **Step 1: Determine F_s(Sampling Frequency) from Hardware Limits**
  - F_s is not determined by the math; it is dictated by your physical op-amp and switch sizing.
  - If your op-amp takes 80 ns to fully settle its output to within ± 0.5 LSB, your clock period must be at least 100 ns(leaving margin).
  - A 100 ns period means your maximum $F_s = 10$ MHz.
- **Step 2: Choose $N_{samples}$ based on DAC Resolution (N)**
  - To accurately measure the noise floor in an FFT, your total number of samples ($N_{samples}$) must be a power of 2, and it must be significantly larger than the total number of distinct digital codes (2^N). A common rule of thumb is $N_{samples} \geq 4 \times 2^N$.
  - **4-bit** (16 codes): Use $N_{samples} = 64$ or 128.
  - **5-bit** (32 codes): Use $N_{samples} = 128$ or 256.
  - **10-bit** (1024 codes): Use $N_{samples} = 4096$ or 8192.

- **Step 3: Choose M(Number of Cycles) to be Mutually Prime**
  - $M$is the number of complete sine wave cycles you capture. It must be an integer.
  - **Crucial Rule:** $M$and $N_{samples}$must be mutually prime (coprime). Because $N_{samples}$is always a power of 2, Mjust needs to be an **odd number**.
  - If you use an even number, the samples will hit the exact same voltage points on every single cycle, masking dynamic DNL errors.
  - Choosing a prime number (e.g., 3, 7, 11, 23) forces the sampling to slightly shift phase on every cycle, thoroughly testing every possible transition in the DAC.

Step 4: Calculate the Target f_{in}

Once you have the physical F_s, the mathematical $N_{samples}$, and an odd prime M, plug them into the equation to find the exact analog frequency you are testing.

Calculation Examples (Assuming a maximum F_s of 10 MHz)

Parameter	4-Bit DAC	5-Bit DAC	10-Bit DAC
Max Codes (2^N)	16	32	1024
Chosen $N_{samples}$	64	128	4096
Chosen F_s	10 MHz	10 MHz	10 MHz
Chosen M(Odd/Prime)	3	5	23
Calculated f_{in}	468.75 kHz	390.62 kHz	56.15 kHz
Calculation Math	$\frac{3}{64} \times 10$ MHz	$\frac{5}{128} \times 10$ MHz	$\frac{23}{4096} \times 10$ MHz
Total Simulation Time	(No. of Samples/F_s) 6.4 us	12.8 us	2.4414ms

- Changing the sampling frequency to 1 MHz means your digital clock period is exactly 1 µs. The DAC will process a new digita word and update its analog output every 1 µs.

- The coherent sampling equation $f_{in} = \frac{M}{N_{samples}} \times F_s$ yields new analog input frequencies and total simulation times.

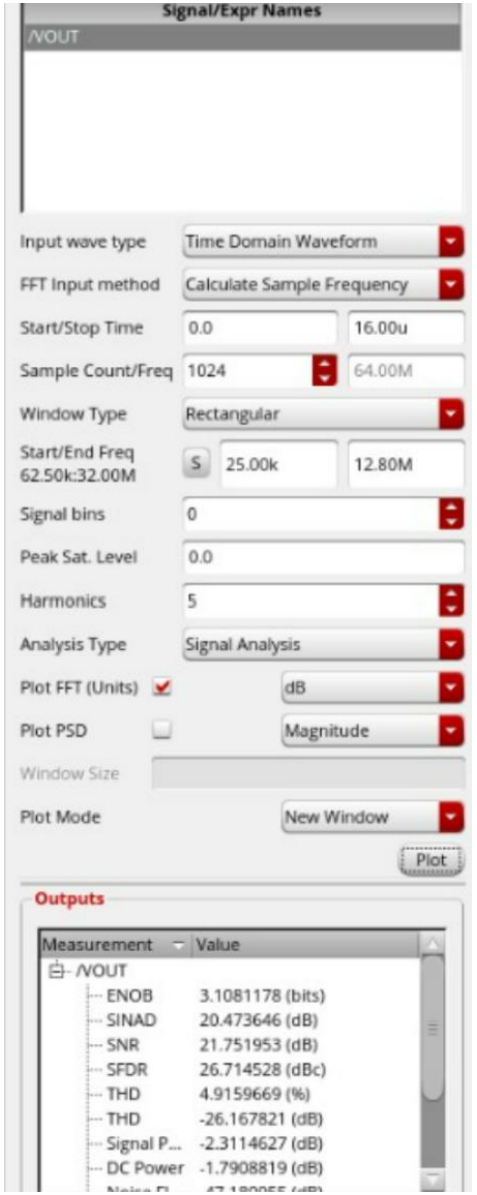
Parameter	4-Bit DAC		5-Bit DAC	10-Bit DAC
Max Codes (2^N)	16		32	1024
Chosen $N_{samples}$	64		128	4096
Chosen F_s	1 MHz	4MHz	1 MHz	1 MHz
Chosen M(Odd/Prime)	3	1	5	23
Calculated f_{in}	46.875 kHz	62.5KHz	39.0625 kHz	5.6152 kHz
Calculation Math	(3 / 64) × 1 MHz	(1/64) x 4 MHz	(5 / 128) × 1 MHz	(23 / 4096) × 1 MHz
Total Simulation Time	64 µs	16µs	128 µs	4096 µs (4.096 ms)

- **For applying input use vpwlf**, which stands for **Voltage Piece-Wise Linear File**.
- In Cadence Virtuoso, vpwlf is an ideal voltage source component found in the standard analogLib library.
- Add four Voltage pulse-wise linear file instances.
- Give path to D0.txt, D1.txt, D2.txt and D3.txt in filename tab.
- Give period as simulation time we calculated
- Run transient Simulation
- Open Spectrum window and Select V_{out} waveform
- Apply settings
- Click plot

SPECTRAL WINDOW SETTINGS

- Based on your parameters ($F_s = 1$ MHz, $N = 64$, $M = 3$), update the fields in your Spectrum window to the following exact values:
- **Start/Stop Time:** 0.0 to 64.00u
  - $F_s = 1$ MHz is exactly 64 us ($T_{sim} = 64/1$ MHz).
- **Sample Count:** 64
  - Exactly 64 discrete data points to the FFT engine.
- **Window Type:** Rectangular
  - Because your sequence perfectly satisfies the coherent sampling equation ($f_{in}/F_s = 3/64$), the waveform seamlessly wraps around.
  - A Rectangular window mathematically preserves this coherence.
- **Harmonics:** 5 (or up to 7)
  - This defines how many integer multiples of your fundamental analog frequency (46.875 kHz) are factored into the Total Harmonic Distortion (THD) and SINAD calculations.
- **Analysis Type:** Signal Analysis





- **Start Freq:** 0 (or 15.625 kHz)
  - The lowest non-zero frequency your FFT can resolve is calculated as F_s/N.
  - For the setup, $1\ \text{MHz}/64 = 15.625\ \text{kHz}$.
- Setting it to 0 includes the DC offset, while setting it to 15.625 kHz skips DC to focus strictly on the AC signals.
- **End Freq:** 500k
  - According to the Nyquist-Shannon sampling theorem, the absolute maximum frequency an FFT can physically represent without aliasing is exactly half of your sampling frequency ($F_s/2$).
  - Since sampling frequency is 1 MHz, your Nyquist limit is 500 kHz.

MAXIMUM SAMPLING FREQUENCY

- The maximum working frequency (sampling rate, F_s) of a DAC is strictly determined by its **worst-case settling time**. You cannot clock the DAC faster than the time it takes for the analog output to physically transition and stabilize to its correct voltage.

- To find the maximum working frequency in Cadence Virtuoso, you must simulate the **Major Carry Transition**, as this causes the largest internal voltage swings and glitch impulses.

MAXIMUM SAMPLING FREQUENCY

Step 3: Measure the Settling Time ($t_{settling}$)

In the Virtuoso ViVA waveform viewer:

1. Place your first cursor (Cursor A) exactly at the 50% voltage threshold of the rising clock edge that triggered the transition.
2. Look at your analog output voltage. It will likely slew, overshoot, and ring.
3. Place your second cursor (Cursor B) at the exact moment the ringing permanently enters the ± 56.25 mVerror band of the target voltage (0.9 V for Code 8).
4. The time difference (ΔX) between Cursor A and Cursor B is your **worst-case settling time**.

Step 4: Calculate the Maximum Frequency

- Once you extract the settling time, the theoretical maximum frequency is simply the inverse:

$$F_{max} = \frac{1}{t_{settling}}$$

- **Determine the Error Band**
  - Before running the simulation, calculate your ± 0.5LSB error band. The DAC is considered "settled" only when the output voltage enters this band and stops ringing.
  - For your 1.8 V, 4-bit DAC, 1 LSB $= 112.5$ mV.
  - The error band is ± 56.25 mVaround the final ideal voltage.
- **Set up the Worst-Case Transient Test**
  - The longest settling time occurs when the maximum number of CMOS switches change state simultaneously. In an R-2R ladder, this is the mid-scale major carry transition.
  - Set up a standard tran simulation.
  - Drive your digital inputs to transition from 0111 (Code 7) to 1000 (Code 8).
  - Set the input clock speed to be very slow (e.g., 1 musperiod) so you can observe the complete, unrestricted settling behavior.

MAXIMUM SAMPLING FREQUENCY

In a real chip tapeout, you never run the DAC at the absolute theoretical limit because PVT (Process, Voltage, Temperature) variations will slow down the silicon. You must add a safety margin (typically 20% to 30%) to the settling time.

$$F_{working} = \frac{1}{t_{settling} \times 1.25}$$

MAXIMUM SAMPLING FREQUENCY

- If your calculated F_{max} is lower than your target specification, the bottleneck is usually one of two components:
  - **The Op-Amp (Dominant):** The I-V conversion op-amp's Slew Rate and Gain-Bandwidth Product (GBW) dictate how fast the output can move. If the waveform looks like a straight diagonal ramp rather than a sharp curve, your op-amp is slew-rate limited.
  - **The R-2R Ladder RC Constant:** The R_{ON} resistance of your CMOS switches interacting with the parasitic capacitance of the resistors creates an intrinsic low-pass filter. Increasing the W/L ratio of your switches lowers R_{ON} and speeds up the RC time constant.

Sample (n)	Formula: 7.5(1+(sin(2π*n)/64))	Float Value	Quantized Code D[n]	Binary	time -	D3	D2	D1	D0
0	7.5·(1+0.0000)	7.5	8	1000	0	0.000u 1.8	0.000u 0.0	0.000u 0.0	0.000u 0.0
1	7.5·(1+0.0980)	8.24	8	1000	2.50E-07	8.249u 1.8	1.499u 0.0	0.749u 0.0	0.499u 0.0
2	7.5·(1+0.1951)	8.96	9	1001	5.00E-07	8.250u 0.0	1.500u 1.8	0.750u 1.8	0.500u 1.8
3	7.5·(1+0.2903)	9.68	10	1010	7.50E-07	16.000u 0.0	6.749u 1.8	1.499u 1.8	0.749u 1.8
4	7.5·(1+0.3827)	10.37	10	1010	1.00E-06		6.750u 0.0	1.500u 0.0	0.750u 0.0
5	7.5·(1+0.4714)	11.04	11	1011	1.25E-06		8.249u 0.0	2.499u 0.0	1.249u 0.0
6	7.5·(1+0.5556)	11.67	12	1100	1.50E-06		8.250u 1.8	2.500u 1.8	1.250u 1.8
7	7.5·(1+0.6344)	12.26	12	1100	1.75E-06		9.499u 1.8	5.749u 1.8	1.499u 1.8
8	7.5·(1+0.7071)	12.8	13	1101	2.00E-06		9.500u 0.0	5.750u 0.0	1.500u 0.0
9	7.5·(1+0.7730)	13.3	13	1101	2.25E-06		14.749u 0.0	6.749u 0.0	1.999u 0.0
10	7.5·(1+0.8315)	13.74	14	1110	2.50E-06		14.750u 1.8	6.750u 1.8	2.000u 1.8
11	7.5·(1+0.8819)	14.11	14	1110	2.75E-06		16.000u 1.8	7.499u 1.8	2.499u 1.8
12	7.5·(1+0.9239)	14.43	14	1110	3.00E-06			7.500u 0.0	2.500u 0.0
13	7.5·(1+0.9569)	14.68	15	1111	3.25E-06			8.249u 0.0	3.249u 0.0
14	7.5·(1+0.9808)	14.86	15	1111	3.50E-06			8.250u 1.8	3.250u 1.8
15	7.5·(1+0.9952)	14.96	15	1111	3.75E-06			8.749u 1.8	4.999u 1.8
16	7.5·(1+1.0000)	15	15	1111	4.00E-06			8.750u 0.0	5.000u 0.0
17	7.5·(1+0.9952)	14.96	15	1111	4.25E-06			9.499u 0.0	5.749u 0.0
18	7.5·(1+0.9808)	14.86	15	1111	4.50E-06			9.500u 1.8	5.750u 1.8
19	7.5·(1+0.9569)	14.68	15	1111	4.75E-06			10.499u 1.8	6.249u 1.8
20	7.5·(1+0.9239)	14.43	14	1110	5.00E-06			10.500u 0.0	6.250u 0.0
21	7.5·(1+0.8819)	14.11	14	1110	5.25E-06			13.749u 0.0	6.749u 0.0
22	7.5·(1+0.8315)	13.74	14	1110	5.50E-06			13.750u 1.8	6.750u 1.8
23	7.5·(1+0.7730)	13.3	13	1101	5.75E-06			14.749u 1.8	6.999u 1.8
24	7.5·(1+0.7071)	12.8	13	1101	6.00E-06			14.750u 0.0	7.000u 0.0
25	7.5·(1+0.6344)	12.26	12	1100	6.25E-06			15.499u 0.0	7.499u 0.0
26	7.5·(1+0.5556)	11.67	12	1100	6.50E-06			15.500u 1.8	7.500u 1.8
27	7.5·(1+0.4714)	11.04	11	1011	6.75E-06			16.000u 1.8	7.749u 1.8
28	7.5·(1+0.3827)	10.37	10	1010	7.00E-06				7.750u 0.0
29	7.5·(1+0.2903)	9.68	10	1010	7.25E-06				8.249u 0.0
30	7.5·(1+0.1951)	8.96	9	1001	7.50E-06				8.250u 1.8
31	7.5·(1+0.0980)	8.24	8	1000	7.75E-06				8.499u 1.8
32	7.5·(1+0.0000)	7.5	8	1000	8.00E-06				8.500u 0.0
33	7.5·(1−0.0980)	6.76	7	111	8.25E-06				8.749u 0.0
34	7.5·(1−0.1951)	6.04	6	110	8.50E-06				8.750u 1.8
35	7.5·(1−0.2903)	5.32	5	101	8.75E-06				9.249u 1.8
36	7.5·(1−0.3827)	4.63	5	101	9.00E-06				9.250u 0.0
37	7.5·(1−0.4714)	3.97	4	100	9.25E-06				9.499u 0.0
38	7.5·(1−0.5556)	3.33	3	11	9.50E-06				9.500u 1.8
39	7.5·(1−0.6344)	2.74	3	11	9.75E-06				9.999u 1.8



40	7.5·(1−0.7071)	2.2	2	10	1.00E-05				10.000u 0.0
41	7.5·(1−0.7730)	1.7	2	10	1.03E-05				10.499u 0.0
42	7.5·(1−0.8315)	1.26	1	1	1.05E-05				10.500u 1.8
43	7.5·(1−0.8819)	0.89	1	1	1.08E-05				11.249u 1.8
44	7.5·(1−0.9239)	0.57	1	1	1.10E-05				11.250u 0.0
45	7.5·(1−0.9569)	0.32	0	0	1.13E-05				12.999u 0.0
46	7.5·(1−0.9808)	0.14	0	0	1.15E-05				13.000u 1.8
47	7.5·(1−0.9952)	0.04	0	0	1.18E-05				13.749u 1.8
48	7.5·(1−1.0000)	0	0	0	1.20E-05				13.750u 0.0
49	7.5·(1−0.9952)	0.04	0	0	1.23E-05				14.249u 0.0
50	7.5·(1−0.9808)	0.14	0	0	1.25E-05				14.250u 1.8
51	7.5·(1−0.9569)	0.32	0	0	1.28E-05				14.749u 1.8
52	7.5·(1−0.9239)	0.57	1	1	1.30E-05				14.750u 0.0
53	7.5·(1−0.8819)	0.89	1	1	1.33E-05				14.999u 0.0
54	7.5·(1−0.8315)	1.26	1	1	1.35E-05				15.000u 1.8
55	7.5·(1−0.7730)	1.7	2	10	1.38E-05				15.499u 1.8
56	7.5·(1−0.7071)	2.2	2	10	1.40E-05				15.500u 0.0
57	7.5·(1−0.6344)	2.74	3	11	1.43E-05				15.749u 0.0
58	7.5·(1−0.5556)	3.33	3	11	1.45E-05				15.750u 1.8
59	7.5·(1−0.4714)	3.97	4	100	1.48E-05				16.000u 1.8
60	7.5·(1−0.3827)	4.63	5	101	1.50E-05				
61	7.5·(1−0.2903)	5.32	5	101	1.53E-05				
62	7.5·(1−0.1951)	6.04	6	110	1.55E-05				
63	7.5·(1−0.0980)	6.76	7	111	1.60E-05				

Sample (n)	Formula: 7.5·(1+sin(2π·(3/64)·n))	Float Value	Quantized Code D[n]	Binary	Time (µs)	D3	D2	D1	D0	D3	D2	D1	D0
0	7.5·(1+sin(0.0000))	7.5	8	1000	0	1	0	0	0	0.000u 1.8	0.000u 0.0	0.000u 0.0	0.000u 0.0
1	7.5·(1+sin(0.2945))	9.68	10	1010	1	1	0	1	0	10.999u 1.8	1.999u 0.0	0.999u 0.0	2.999u 0.0
2	7.5·(1+sin(0.5890))	11.67	12	1100	2	1	1	0	0	11.000u 0.0	2.000u 1.8	1.000u 1.8	3.000u 1.8
3	7.5·(1+sin(0.8836))	13.3	13	1101	3	1	1	0	1	21.999u 0.0	8.999u 1.8	1.999u 1.8	3.999u 1.8
4	7.5·(1+sin(1.1781))	14.43	14	1110	4	1	1	1	0	22.000u 1.8	9.000u 0.0	2.000u 0.0	4.000u 0.0
5	7.5·(1+sin(1.4726))	14.96	15	1111	5	1	1	1	1	32.999u 1.8	10.999u 0.0	2.999u 0.0	4.999u 0.0
6	7.5·(1+sin(1.7671))	14.86	15	1111	6	1	1	1	1	33.000u 0.0	11.000u 1.8	3.000u 1.8	5.000u 1.8
7	7.5·(1+sin(2.0617))	14.11	14	1110	7	1	1	1	0	42.999u 0.0	12.999u 1.8	9.999u 1.8	6.999u 1.8
8	7.5·(1+sin(2.3562))	12.8	13	1101	8	1	1	0	1	43.000u 1.8	13.000u 0.0	10.000u 0.0	7.000u 0.0
9	7.5·(1+sin(2.6507))	11.04	11	1011	9	1	0	1	1	53.999u 1.8	19.999u 0.0	10.999u 0.0	7.999u 0.0
10	7.5·(1+sin(2.9452))	8.96	9	1001	10	1	0	0	1	54.000u 0.0	20.000u 1.8	11.000u 1.8	8.000u 1.8
11	7.5·(1+sin(3.2398))	6.76	7	111	11	0	1	1	1	64.000u 0.0	21.999u 1.8	11.999u 1.8	14.999u 1.8
12	7.5·(1+sin(3.5343))	4.63	5	101	12	0	1	0	1		22.000u 0.0	12.000u 0.0	15.000u 0.0
13	7.5·(1+sin(3.8288))	2.74	3	11	13	0	0	1	1		23.999u 0.0	12.999u 0.0	17.999u 0.0
14	7.5·(1+sin(4.1233))	1.26	1	1	14	0	0	0	1		24.000u 1.8	13.000u 1.8	18.000u 1.8
15	7.5·(1+sin(4.4179))	0.32	0	0	15	0	0	0	0		30.999u 1.8	13.999u 1.8	24.999u 1.8
16	7.5·(1+sin(4.7124))	0	0	0	16	0	0	0	0		31.000u 0.0	14.000u 0.0	25.000u 0.0
17	7.5·(1+sin(5.0069))	0.32	0	0	17	0	0	0	0		32.999u 0.0	18.999u 0.0	25.999u 0.0
18	7.5·(1+sin(5.3014))	1.26	1	1	18	0	0	0	1		33.000u 1.8	19.000u 1.8	26.000u 1.8
19	7.5·(1+sin(5.5960))	2.74	3	11	19	0	0	1	1		33.999u 1.8	19.999u 1.8	27.999u 1.8
20	7.5·(1+sin(5.8905))	4.63	5	101	20	0	1	0	1		34.000u 0.0	20.000u 0.0	28.000u 0.0
21	7.5·(1+sin(6.1850))	6.76	7	111	21	0	1	1	1		40.999u 0.0	20.999u 0.0	28.999u 0.0
22	7.5·(1+sin(6.4795))	8.96	9	1001	22	1	0	0	1		41.000u 1.8	21.000u 1.8	29.000u 1.8
23	7.5·(1+sin(6.7741))	11.04	11	1011	23	1	0	1	1		42.999u 1.8	21.999u 1.8	29.999u 1.8
24	7.5·(1+sin(7.0686))	12.8	13	1101	24	1	1	0	1		43.000u 0.0	22.000u 0.0	30.000u 0.0
25	7.5·(1+sin(7.3631))	14.11	14	1110	25	1	1	1	0		44.999u 0.0	22.999u 0.0	32.999u 0.0
26	7.5·(1+sin(7.6576))	14.86	15	1111	26	1	1	1	1		45.000u 1.8	23.000u 1.8	33.000u 1.8
27	7.5·(1+sin(7.9522))	14.96	15	1111	27	1	1	1	1		51.999u 1.8	29.999u 1.8	34.999u 1.8
28	7.5·(1+sin(8.2467))	14.43	14	1110	28	1	1	1	0		52.000u 0.0	30.000u 0.0	35.000u 0.0
29	7.5·(1+sin(8.5412))	13.3	13	1101	29	1	1	0	1		53.999u 0.0	30.999u 0.0	35.999u 0.0
30	7.5·(1+sin(8.8357))	11.67	12	1100	30	1	1	0	0		54.000u 1.8	31.000u 1.8	36.000u 1.8
31	7.5·(1+sin(9.1303))	9.68	10	1010	31	1	0	1	0		55.999u 1.8	31.999u 1.8	36.999u 1.8
32	7.5·(1+sin(9.4248))	7.5	8	1000	32	1	0	0	0		56.000u 0.0	32.000u 0.0	37.000u 0.0
33	7.5·(1+sin(9.7193))	5.32	5	101	33	0	1	0	1		62.999u 0.0	33.999u 0.0	38.999u 0.0
34	7.5·(1+sin(10.0138))	3.33	3	11	34	0	0	1	1		63.000u 1.8	34.000u 1.8	39.000u 1.8
35	7.5·(1+sin(10.3084))	1.7	2	10	35	0	0	1	0		64.000u 1.8	35.999u 1.8	39.999u 1.8
36	7.5·(1+sin(10.6029))	0.57	1	1	36	0	0	0	1			36.000u 0.0	40.000u 0.0
37	7.5·(1+sin(10.8974))	0.04	0	0	37	0	0	0	0			39.999u 0.0	46.999u 0.0
38	7.5·(1+sin(11.1919))	0.14	0	0	38	0	0	0	0			40.000u 1.8	47.000u 1.8
39	7.5·(1+sin(11.4865))	0.89	1	1	39	0	0	0	1			40.999u 1.8	49.999u 1.8
40	7.5·(1+sin(11.7810))	2.2	2	10	40	0	0	1	0			41.000u 0.0	50.000u 0.0
41	7.5·(1+sin(12.0755))	3.97	4	100	41	0	1	0	0			41.999u 0.0	56.999u 0.0
42	7.5·(1+sin(12.3700))	6.04	6	110	42	0	1	1	0			42.000u 1.8	57.000u 1.8
43	7.5·(1+sin(12.6646))	8.24	8	1000	43	1	0	0	0			42.999u 1.8	57.999u 1.8
44	7.5·(1+sin(12.9591))	10.37	10	1010	44	1	0	1	0			43.000u 0.0	58.000u 0.0
45	7.5·(1+sin(13.2536))	12.26	12	1100	45	1	1	0	0			43.999u 0.0	59.999u 0.0
46	7.5·(1+sin(13.5481))	13.74	14	1110	46	1	1	1	0			44.000u 1.8	60.000u 1.8
47	7.5·(1+sin(13.8427))	14.68	15	1111	47	1	1	1	1			44.999u 1.8	60.999u 1.8
48	7.5·(1+sin(14.1372))	15	15	1111	48	1	1	1	1			45.000u 0.0	61.000u 0.0
49	7.5·(1+sin(14.4317))	14.68	15	1111	49	1	1	1	1			45.999u 0.0	61.999u 0.0
50	7.5·(1+sin(14.7262))	13.74	14	1110	50	1	1	1	0			46.000u 1.8	62.000u 1.8
51	7.5·(1+sin(15.0208))	12.26	12	1100	51	1	1	0	0			50.999u 1.8	64.000u 1.8
52	7.5·(1+sin(15.3153))	10.37	10	1010	52	1	0	1	0			51.000u 0.0	
53	7.5·(1+sin(15.6098))	8.24	8	1000	53	1	0	0	0			51.999u 0.0	
54	7.5·(1+sin(15.9043))	6.04	6	110	54	0	1	1	0			52.000u 1.8	
55	7.5·(1+sin(16.1989))	3.97	4	100	55	0	1	0	0			52.999u 1.8	
56	7.5·(1+sin(16.4934))	2.2	2	10	56	0	0	1	0			53.000u 0.0	
57	7.5·(1+sin(16.7879))	0.89	1	1	57	0	0	0	1			53.999u 0.0	
58	7.5·(1+sin(17.0824))	0.14	0	0	58	0	0	0	0			54.000u 1.8	
59	7.5·(1+sin(17.3770))	0.04	0	0	59	0	0	0	0			54.999u 1.8	
60	7.5·(1+sin(17.6715))	0.57	1	1	60	0	0	0	1			55.000u 0.0	
61	7.5·(1+sin(17.9660))	1.7	2	10	61	0	0	1	0			55.999u 0.0	
62	7.5·(1+sin(18.2605))	3.33	3	11	62	0	0	1	1			56.000u 1.8	
63	7.5·(1+sin(18.5551))	5.32	5	101	63	0	1	0	1			56.999u 1.8	
												57.000u 0.0	
												60.999u 0.0	
												61.000u 1.8	
												62.999u 1.8	
												63.000u 0.0	
												64.000u 0.0	

